



Data Handling: Import, Cleaning and Visualisation

Lecture 7:

Data Preparation, Manipulation, and Cleaning I

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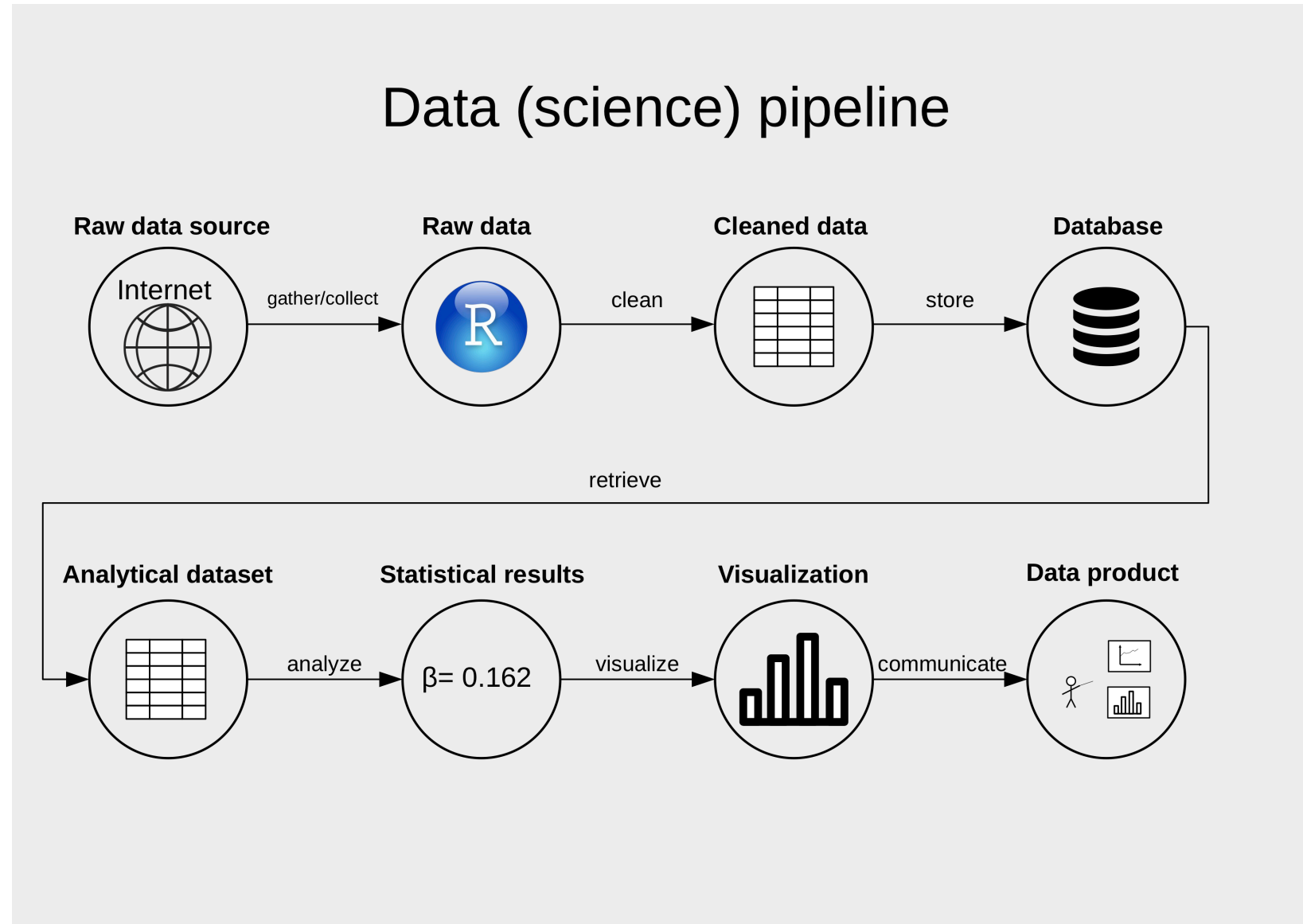
2025-11-13

Welcome back!

Recap of Part I



Part I: Data, data processing, data structures



Part I: Data, data processing, data structures

Our story so far:

-  Computers store data as bits (Bits → Bytes → Encoded Text)
-  Text encodes meaning
-  Files encode structure (→ Structured Files like CSV, JSON, XML)
-  Structure is more or less flexible (→ rectangular data, hierarchical data)
-   Structure is obtained automatically or manually (→ APIs vs local files)
-  R parses structure into memory. (→ R Parsing → R Data Structures)

Part I: Your own Group Project



Part I: Your own Group Project



Recap: non-rectangular data

XML vs JSON: Structure and Syntax

XML:

```
<person>
  <firstName>John</firstName>
  <lastName>Smith</lastName>
  <age>25</age>
  <address>
    <streetAddress>21 2nd Street</streetAddress>
    <city>New York</city>
    <state>NY</state>
    <postalCode>10021</postalCode>
  </address>
  <phoneNumber>
    <type>home</type>
    <number>212 555-1234</number>
  </phoneNumber>
  <phoneNumber>
    <type>fax</type>
    <number>646 555-4567</number>
  </phoneNumber>
</person>
```

- **Tags:** `<element>value</element>` or `<element attribute="value"/>`
- **Repetition:** Multiple `<phoneNumber>` elements
- **Verbose:** More characters for structure

JSON:

```
{ "firstName": "John",
  "lastName": "Smith",
  "age": 25,
  "address": {
    "streetAddress": "21 2nd Street",
    "city": "New York",
    "state": "NY",
    "postalCode": "10021"
  },
  "phoneNumber": [
    {
      "type": "home",
      "number": "212 555-1234"
    },
    {
      "type": "fax",
      "number": "646 555-4567"
    }
  ]
}
```

- **Objects:** `{ }` for key-value pairs
- **Arrays:** `[ ]` for sequences
- **Compact:** Less verbose syntax

Working with XML and JSON in R

XML in R:

```
library(xml2)

# Read XML file
xml_data <- read_xml("data/person.xml")

# Extract specific elements with Xpath
name <- xml_text(xml_find_all(xml_data, ".//firstName"))
phone_numbers <- xml_text(xml_find_all(xml_data, ".//phoneNumber"))
phone_numbers <- xml_text(xml_children(xml_children(xml_data)))

# Convert to data frame
person_df <- data.frame(
  firstName = name,
  phone = c(phone_numbers[1], phone_numbers[2])
)
```

	firstName	phone
1	John	212 555-1234
2	John	646 555-4567

JSON in R:

```
library(jsonlite)

# Read JSON file
json_doc <- fromJSON("data/person.json")

# Access elements directly
name <- json_doc$firstName
phones <- json_doc$phoneNumber$number

# Convert to data frame (automatic!)
person_df <- as.data.frame(json_doc)
```

	firstName	lastName	age	address.streetAddress	address.city	address.state	address.postalCode	
1	John	Smith	25	21 2nd Street	New York	NY	10021	New
2	John	Smith	25	21 2nd Street	New York	NY	10021	New

	phoneNumber.type	phoneNumber.number	type
1	home	212 555-1234	male
2	fax	646 555-4567	male

HTML: Semi-structured Web Data

HTML documents/webpages consist of '*semi-structured data*'.

- A webpage can contain HTML tables (*structured data*)...
- ...but likely also contains just raw text (*unstructured data*).

Web scraping

- Extract data from HTML using XPath
- Example in class: parser for a Wikipedia table (webscraper) with **rvest**

Raw HTML structure:

```
<table class="wikitable">
<tbody>
<tr>
  <th>Year</th>
  <th>GDP (billions of
    <a href="/wiki/Swiss_franc">CHF</a>)
  </th>
  <th>US dollar exchange</th>
</tr>
<tr>
  <td>1980</td>
  <td>184</td>
  <td>1.67 Francs</td>
</tr>
...
</tbody>
</table>
```


REST APIs: Programmatic Data Access

REST APIs

- **Structured data:** APIs return data in JSON/XML format
- **HTTP requests:** Use GET, POST, PUT, DELETE methods
- **Real-time data:** Access current information (stock prices, weather, economic indicators)
- **(Authentication:** Often require API keys or tokens)
- **Rate limiting:** Restrictions on number of requests per time period

In R: Use **httr2**, **jsonlite** packages to consume APIs

Your group project

Use APIs to retrieve World Bank Data and OECD Data.

Warm-up

JSON files: open-ended question

You import the following json file using `fromJSON()` and store the file in the object `students` in R, i.e. `students <- fromJSON("students.json")`.

```
{
  "students": [
    {
      "id": 19091,
      "firstName": "Peter",
      "lastName": "Mueller",
      "grades": {
        "micro": 5,
        "macro": 4.5,
        "data handling": 5.5
      }
    },
    {
      "id": 19092,
      "firstName": "Anna",
      "lastName": "Schmid",
      "grades": {
        "micro": 5.25,
        "macro": 4,
        "data handling": 5.75
      }
    },
    {
```

- The object `students` is a list.
- `students$students$firstName[2]` returns “Anna”
- In JSON, the value for the “students” key is itself an object (key-value pair).
- In JSON, you have two ways of writing data: arrays and objects. Objects are key-value pairs, and arrays are ordered lists.

```
{  
  "name": "Anna",           // a key "name" with a string value "Anna"  
  "grades": [5, 5.5, 6]     // a key "grades" with an array value [5, 5.5, 6]  
}
```

XML

```
<students>
  <student>
    <id>19091</id>
    <firstName>Peter</firstName>
    <lastName>Mueller</lastName>
    <grades>
      <micro>5</micro>
      <macro>4.5</macro>
      <dataHandling>5.5</dataHandling>
    </grades>
  </student>
  <student>
    <id>19092</id>
    <firstName>Anna</firstName>
    <lastName>Schmid</lastName>
    <grades>
      <micro>5.25</micro>
      <macro>4</macro>
      <dataHandling>5.75</dataHandling>
    </grades>
  </student>
  <student>
    <id>19093</id>
```

- 'students' is the root-node, 'grades' are its children
- The siblings of Trevor Noah are Anna Schmid and Peter Mueller
- This code would be an alternative, equivalent notation for the third student in the xml file above.


```
<student id="19093" firstName="Trevor" lastName="Noah">  
  <grades micro="4" macro="4.5" dataHandling="5" />  
</student>
```

Part II: Data preparation, analysis, and visualization

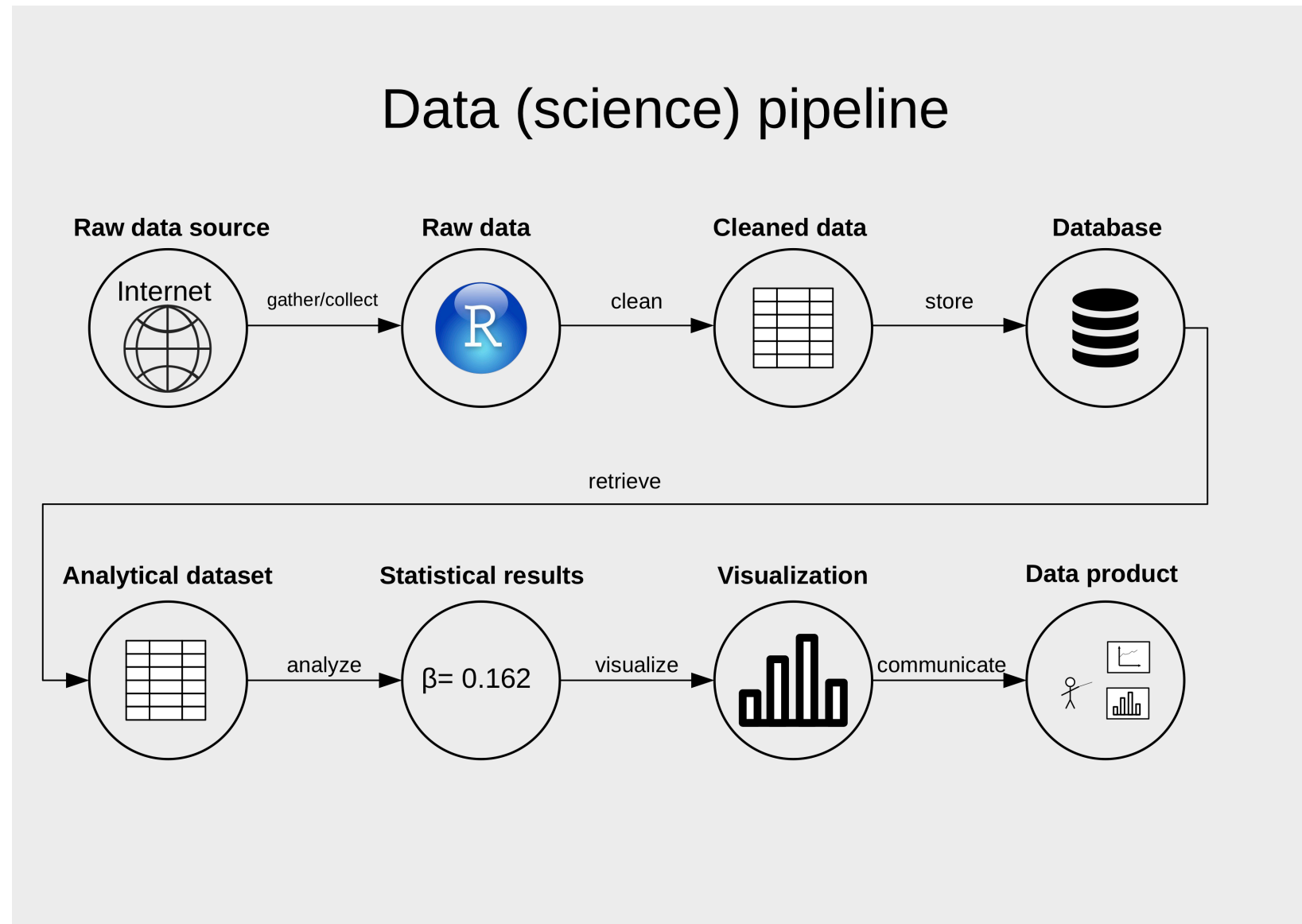
The dataset is imported, now what?

- In practice: still a long way to go.
- Parsable, but messy data: inconsistencies, data types, missing observations, wide format.

The dataset is imported, now what?

- In practice: still a long way to go.
- Parsable, but messy data: inconsistencies, data types, missing observations, wide format.
- **Goal** of data preparation: dataset is ready for analysis.

The dataset is imported, now what?



Part II: Schedule & Organization

Date	Topic
13.11.2025	Data preparation, manipulation, and cleaning I
20.11.2025	Data preparation, manipulation, and cleaning II
20.11.2025	Exercises/Workshop 4: Data gathering, data import
27.11.2025	Guest Lecture: Data Handling @Bank for International Settlements (Andrew Li, Ex-MiQeF, Macroeconomic Analyst)

Part II: Schedule & Organization

Date	Topic
04.12.2025	Visualisation
04.12.2025	Exercises/Workshop 5: Data preparation and applied data analysis with R
11.12.2025	Analytics, more visualisation, and data products
11.12.2025	Short exam on the group project (in class)
18.12.2025	Summary, Wrap-up, Final workshop
18.12.2025	Exercises/Workshop 6: Visualization, dynamic documents
18.12.2025	Exam for Exchange Students

Part II: Schedule & Organization

Exam for exchange students 📅

- 18.12.2024 at 16:15 in room 09-112 (library building).

Walk-ins for the digital exam:

- Wednesday, 3 December 2025 from 10:30 to 14:00 in Room 23-104 (A 23 Lehr-Pavillon)
- Tuesday, 16 December 2025 from 10:30 to 14:00 in 23-203 (A 23 Lehr-Pavillon)

Second part of the project: online today.

- Recommendation: wait for week 9 to start, as we will cover the necessary concepts in class until then.

The Tidyverse

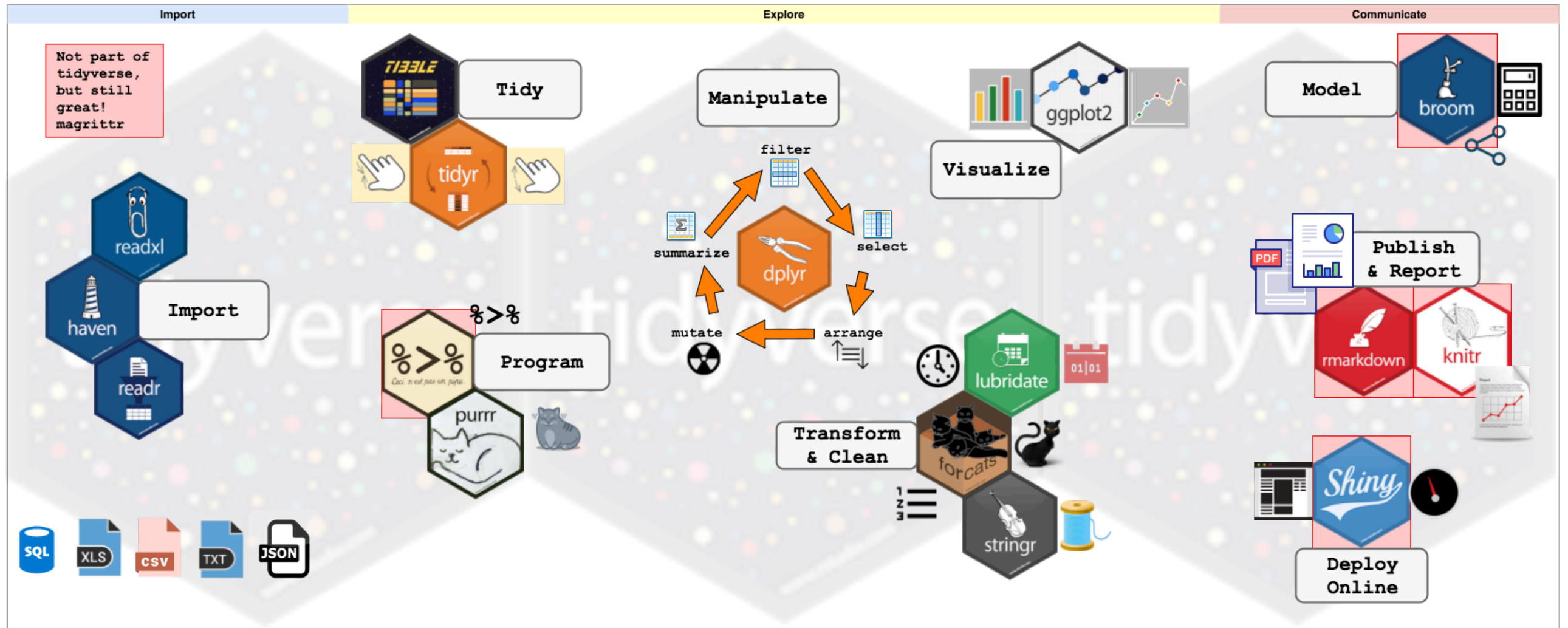
Our main tool in part II: the tidyverse

“The tidyverse is a collection of open source packages for the R programming language introduced by Hadley Wickham and his team that share an underlying design philosophy, grammar, and data structures” of tidy data.” (Wikipedia)

In this course, we will use tidyverse AND base R.



Tidyverse for data handling in R



Source: <https://www.storybench.org/wp-content/uploads/2017/05/tidyverse.png>

Rectangular data in R and tidyverse

- data frames: base R
- tibbles: tidyverse

A note on tibbles and data frames

Tibbles are a modern version of data frames

Similar!

- Tibbles are used in the tidyverse and ggplot2 packages.
- Same information as a data frame.
- Small differences in the manipulation and representation of data.
- See [Tibble vs. DataFrame](#) for more details.

Tibbles are a modern version of data frames

```
Fertility Agriculture Examination Education Catholic Infant.Mortality
Courtelary      80.2      17.0      15      12      9.96      22.2
Delemont        83.1      45.1       6       9     84.84      22.2
Franches-Mnt    92.5      39.7       5       5     93.40      20.2
Moutier         85.8      36.5      12       7     33.77      20.3
Neuveville      76.9      43.5      17      15      5.16      20.6
Porrentruy      76.1      35.3       9       7     90.57      26.6
Broye           83.8      70.2      16       7     92.85      23.6
Glane           92.4      67.8      14       8     97.16      24.9
Gruyere         82.4      53.3      12       7     97.67      21.0
Sarine          82.9      45.2      16      13     91.38      24.4
Veveyse         87.1      64.5      14       6     98.61      24.5
Aigle           64.1      62.0      21      12      8.52      16.5
Aubonne         66.9      67.5      14       7      2.27      19.1
Avenches        68.9      60.7      19      12      4.43      22.7
Cossonay        61.7      69.3      22       5      2.82      18.7
Echallens       68.3      72.6      18       2     24.20      21.2
Grandson        71.7      34.0      17       8      3.30      20.0
Lausanne        55.7      19.4      26      28     12.11      20.2
La Vallee       54.3      15.2      31      20      2.15      10.8
Lavaux          65.1      73.0      19       9      2.84      20.0
Morges           65.5      59.8      22      10      5.23      18.0
Moudon          65.0      55.1      14       3      4.52      22.4
```

Tibbles are a modern version of data frames

```
# A tibble: 47 × 6
  Fertility Agriculture Examination Education Catholic Infant.Mortality
    <dbl>      <dbl>      <int>      <int>      <dbl>      <dbl>
1     80.2         17         15         12         9.96         22.2
2     83.1         45.1          6          9        84.8         22.2
3     92.5         39.7          5          5        93.4         20.2
4     85.8         36.5         12          7        33.8         20.3
5     76.9         43.5         17         15         5.16         20.6
6     76.1         35.3          9          7        90.6         26.6
7     83.8         70.2         16          7        92.8         23.6
8     92.4         67.8         14          8        97.2         24.9
9     82.4         53.3         12          7        97.7          21
10    82.9         45.2         16         13        91.4         24.4
# i 37 more rows
```

Data cleaning

Garbage in, garbage ... ?



Beware of the “Garbage in garbage out” problem

Key conditions:

1. Data values are consistent/clean within each variable.
2. Variables are of proper data types.
3. Dataset is in 'tidy' (long) format.



Data preparation can be understood as consisting of five main actions

- Tidy data.
- **Reshape** datasets from wide to long (and vice versa).
- **Bind** or stack rows in datasets.
- **Join** datasets.
- **Clean** data.

Tidy data: some vocabulary

Following R4DS, a tidy dataset is tidy when...

1. Each **variable** is a **column**; each column is a variable.
2. Each **observation** is a **row**; each row is an observation.
3. Each **value** is a **cell**; each cell is a single value.

Tidy data

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174504898
China	1999	212258	127291272
China	2000	213766	128042583

variables

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174504898
China	1999	212258	127291272
China	2000	213766	128042583

observations

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174504898
China	1999	212258	127291272
China	2000	213766	128042583

values

Tidy data. Source R4DS.

Tidy data is not trivial

In Economics, the definition of an *observation* can vary:

- Panel data (longitudinal data)
- Cross-sectional data
- Time series

Tidy data is not trivial

Panel Data (Longitudinal Data):

- Panel data tracks the same units over multiple time periods: we observe each unit repeatedly across time.
- **Observation:** One unit at one specific time point (e.g., Person A in 2020, Person A in 2021).

Cross-Sectional Data:

- “Snapshot” of different units at the same moment.
- **Observation:** One unit at a single point in time (e.g., Person A, Person B, Person C all measured in 2024).

Time Series Data:

Three examples of non-tidy data (1)

Messy 🤮

```
# A tibble: 2 × 4
  measure `Jan 1` `Jan 2` `Jan 3`
  <chr>    <dbl>    <dbl>    <dbl>
1 Temperature    20      22      21
2 Humidity       80      78      82
```

Tidy 😎

...

Three examples of non-tidy data (1)

Messy 🤮

```
# A tibble: 2 × 4
  measure `Jan 1` `Jan 2` `Jan 3`
  <chr>    <dbl>    <dbl>    <dbl>
1 Temperature    20      22      21
2 Humidity       80      78      82
```

Tidy 😎

```
# A tibble: 3 × 3
  Date      Temperature Humidity
  <chr>        <dbl>    <dbl>
1 Jan 1         20      80
2 Jan 2         22      78
3 Jan 3         21      82
```

Three examples of non-tidy data (2)

Messy 🤮

```
# A tibble: 3 × 2
  year temperature_location
<dbl> <chr>
1  2019 22C_London
2  2019 18C_Paris
3  2019 25C_Rome
```

Tidy 😎

```
homework..
```

Three examples of non-tidy data (3)

Messy 🦌

```
Student Econ DataHandling Management
1 Johannes 5.00          4.0          5.5
2 Hannah 5.25          4.5          6.0
3 Igor 4.00          5.0          6.0
```

Tidy 😎

```
homework..
```

Reshaping

Reshaping: the concept

“Long” format

country	year	metric
x	1960	10
x	1970	13
x	2010	15
y	1960	20
y	1970	23
y	2010	25
z	1960	30
z	1970	33
z	2010	35

“Wide” format

country	yr1960	yr1970	yr2010
x	10	13	15
y	20	23	25
z	30	33	35

Long and wide data. Source: [Hugo Tavares](#)

Wide vs Long Format: Key Differences

Long Format

- **Structure:** Each measurement = one row
- **Variables:** Stacked vertically
- **Dimensions:** More rows, fewer columns
- **Analysis:** Better for statistical operations
- **Example:** Database-normalized format

```
# A tibble: 9 × 3
  Student Subject Grade
  <chr>   <chr>   <dbl>
1 Johannes Econ      5
2 Johannes DataHandling 4
3 Johannes Management 5.5
4 Hannah Econ      5.25
5 Hannah DataHandling 4.5
6 Hannah Management 6
7 Igor Econ      4
8 Igor DataHandling 5
9 Igor Management 6
```

Wide Format

- **Structure:** Each observation = one row
- **Variables:** Spread across multiple columns
- **Dimensions:** More columns, fewer rows
- **Analysis:** Easy for humans to read
- **Example:** Spreadsheet-style layout

	Student	Econ	DataHandling	Management
1	Johannes	5.00	4.0	5.5
2	Hannah	5.25	4.5	6.0
3	Igor	4.00	5.0	6.0

Reshaping: implementation in R

- From wide to long: `melt()`, `gather()`,
 We'll use `tidyverse::pivot_longer()`.
- From long to wide: `cast()`, `spread()`,
 We'll use `tidyverse::pivot_wider()`.

Reshaping: implementation in R with `tidyverse()`

country	year	metric
x	1960	10
x	1970	13
x	2010	15
y	1960	20
y	1970	23
y	2010	25
z	1960	30
z	1970	33
z	2010	35

`pivot_wider(names_from = "year",
names_prefix = "yr",
values_from = "metric")`

country	yr1960	yr1970	yr2010
x	10	13	15
y	20	23	25
z	30	33	35

`pivot_longer(cols = yr1960:yr2010,
names_to = "year",
names_prefix = "yr",
values_to = "metric")`

When to Use Wide vs Long Format

Use Long Format for:

- **Data analysis** with tidyverse/ggplot2
- **Grouping and summarizing operations** (summarize, mutate by groups)
- **Time series analysis**
- **Following tidy data principles**
- **Creating visualizations** with multiple variables

Use Wide Format for:

- **Data entry** and manual correction
- **Reports and tables** for presentation
- **Statistical models** like regressions where each predictor should be a separate column (**lm**), many machine learning functions
- **Spreadsheet analysis** (Excel-style operations)
- **Human readability** and quick scanning

Rule of thumb: Wide format for presentation, long format for analysis

Stack and row-bind

Stack/row-bind: the concept

ID	X	Y
1	a	50
2	b	10

ID	Z
3	M
4	O

ID	X	Z
5	c	P

ID	X	Y	Z
1	a	50	NA
2	b	10	NA
3	NA	NA	M
4	NA	NA	O
5	c	NA	P

Stack/row-bind: implementation in R

- Use `rbind()` in base R
 - Requires that the data frames have the same column names and same column classes.
- Use `bind_rows()` from `dplyr()`
 - More flexible
 - Binds data frames with different column names and classes
 - Automatically fills missing columns with `NA`

For these reasons (+ performance, handling of row names, and handling of factors), `dplyr::bind_rows()` is preferred in most applications.

Stack/row-bind: code example

```
# Create three dfs
subset1 <- data.frame(ID = c(1,2),
                      X = c("a", "b"),
                      Y = c(50,10))

subset2 <- data.frame(ID = c(3,4),
                      Z = c("M", "O"))

subset3 <- data.frame(ID = c(5),
                      X = c("c"),
                      Z = "P")

# Inspect
subset1
```

```
  ID X  Y
1  1 a 50
2  2 b 10
```

```
subset2
```

```
  ID Z
1  3 M
2  4 O
```

```
subset3
```

	ID	X	Z
1	5	c	P

Stack/row-bind: code example

```
# Stack data frames
combined_df_bind_rows <- bind_rows(subset1, subset2, subset3)
combined_df_rbind      <- rbind(subset1, subset2, subset3)
```

```
Error in rbind(deparse.level, ...): numbers of columns of arguments do not match
```

```
# What are the following objects?
combined_df_bind_rows
```

	ID	X	Y	Z
1	1	a	50	<NA>
2	2	b	10	<NA>
3	3	<NA>	NA	M
4	4	<NA>	NA	O
5	5	c	NA	P

```
combined_df_rbind
```

```
Error: object 'combined_df_rbind' not found
```

Appendix

Tibbles vs data frames

- Unlike data frames, tibbles don't show the entire dataset when you print it
- Tibbles cannot access a column when you provide a partial name of the column, but data frames can.
- When you access only one column of a tibble, it will keep the tibble structure. But when you access one column of a data frame, it will become a vector.
- When assigning a new column to a tibble, the input will not be recycled, which means you have to provide an input of the same length of the other columns. But a data frame will recycle the input.
- Tibbles preserve all the variable types, while data frames have the option to convert string into factor. (In older versions of R, data frames will convert string into factor by default)